

BELFAST INTL, United KingdomWMO: **039170**Lat: **54.6638N**Lon: **6.2251W**Elev: **82**StdP: **100.35**Time Zone: **0.00 (EUW)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.9	-1.3	-5.1	2.5	-0.8	-3.7	2.8	0.6	14.2	7.2	12.9	7.8	1.7	70	0.637

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.8	22.6	17.0	20.8	16.4	19.5	15.6	18.0	21.0	17.1	19.8	16.3	18.7	4.0	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.8	12.1	19.2	16.0	11.5	18.4	15.2	10.9	17.6	51.1	21.1	48.3	19.8	46.0	18.7	22.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.4	10.1	9.0	-5.9	25.3	2.5	1.8	-7.8	26.6	-9.2	27.6	-10.6	28.6	-12.4	29.9
			-6.3	19.7	2.4	1.2	-8.0	20.6	-9.4	21.3	-10.8	21.9	-12.5	22.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.6	4.8	5.0	6.2	8.3	10.9	13.6	15.2	15.0	13.2	10.3	7.1	5.2
	DBStd	4.61	2.76	2.70	2.59	2.60	2.62	2.34	2.18	1.93	2.09	2.54	2.80	3.30
	HDD10.0	781	162	142	121	64	19	1	0	0	2	27	92	152
	HDD18.3	3199	419	374	377	302	231	144	99	104	156	248	337	409
	CDD10.0	631	1	1	2	12	47	108	163	156	97	37	6	2
	CDD18.3	7	0	0	0	0	0	1	4	2	0	0	0	0
	CDH23.3	31	0	0	0	0	1	8	16	6	0	0	0	0
	CDH26.7	2	0	0	0	0	0	1	1	0	0	0	0	0
	WSAvg	4.3	4.9	5.0	4.7	4.4	4.2	3.9	3.7	3.7	3.9	4.5	4.5	4.6
	PrecAvg	856	82	59	63	55	59	63	72	81	76	87	83	79
(15) Precipitation	PrecMax	1071	151	132	117	117	131	155	146	188	163	176	158	195
	PrecMin	693	22	6	20	14	7	25	14	12	4	30	33	14
	PrecStd	84	31	27	23	27	29	27	28	36	39	37	30	34
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	12.3	12.7	14.4	18.2	22.3	24.1	25.5	24.0	20.6	17.2	14.5	13.2	
	0.4% MCWB	11.0	10.8	10.6	12.7	15.9	16.5	18.6	18.0	17.0	14.8	13.3	12.0	
	2% DB	10.9	11.3	12.5	15.7	19.5	21.6	22.8	21.2	19.0	15.8	13.2	11.9	
	2% MCWB	9.9	9.8	10.0	11.5	14.3	16.1	17.3	17.2	16.0	14.0	12.2	10.8	
	5% DB	10.0	10.3	11.3	14.0	17.4	19.6	20.9	19.6	17.8	15.0	12.2	10.9	
	5% MCWB	9.1	9.1	9.3	10.7	13.1	15.0	16.6	16.3	15.0	13.4	11.2	9.9	
	10% DB	9.0	9.2	10.3	12.5	15.5	17.9	19.3	18.4	16.7	14.2	11.3	9.8	
	10% MCWB	8.1	8.0	8.6	9.7	12.0	14.0	15.6	15.5	14.2	12.6	10.3	8.9	
	0.4% WB	11.2	11.2	11.6	13.6	16.6	18.2	19.6	19.3	18.0	15.6	13.7	12.2	
	0.4% MCDB	11.9	12.2	13.3	16.2	21.2	22.6	24.0	22.5	19.8	16.4	14.2	13.1	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	10.1	10.1	10.6	12.2	14.8	16.7	18.1	17.8	16.6	14.5	12.4	11.0	
	2% MCDB	10.7	11.0	12.0	15.0	18.4	20.3	21.5	20.2	18.3	15.5	13.0	11.7	
	5% WB	9.2	9.2	9.7	11.0	13.7	15.6	17.1	16.9	15.6	13.7	11.4	10.1	
	5% MCDB	9.9	10.2	11.0	13.4	16.8	18.7	20.1	19.0	17.2	14.8	12.1	10.8	
	10% WB	8.2	8.2	8.8	10.0	12.6	14.6	16.2	16.1	14.7	12.9	10.4	9.0	
	10% MCDB	9.0	9.2	10.1	12.2	15.1	17.4	18.7	18.0	16.2	14.0	11.2	9.8	
(35) Mean Daily Temperature Range	MDBR	4.6	5.2	6.2	6.9	7.6	7.1	6.8	6.5	6.2	5.4	4.8	4.7	
	5% DB	5.1	5.6	6.9	8.6	10.2	10.0	9.3	8.0	7.3	5.7	5.0	5.5	
	5% WB	4.9	5.0	5.0	5.4	6.1	5.8	5.3	4.9	4.8	4.5	4.8	5.4	
	MCDBR	5.3	5.3	6.2	7.7	9.4	8.8	8.1	7.0	6.3	5.2	4.9	5.5	
	MCWBR	5.2	4.9	5.1	5.3	6.1	5.7	5.1	4.8	4.8	4.6	4.9	5.4	
(40) Clear-Sky Solar Irradiance	taub	0.331	0.342	0.366	0.379	0.384	0.376	0.378	0.362	0.356	0.341	0.325	0.320	
	taud	2.239	2.321	2.295	2.293	2.323	2.374	2.373	2.432	2.446	2.426	2.303	2.204	
	Ebn at Noon	586	734	802	844	861	870	861	855	811	731	595	506	
	Edh at Noon	61	79	100	115	118	113	111	99	86	70	56	51	
(44) All-Sky Solar Radiation	RadAvg	0.57	1.25	2.32	3.75	4.77	4.84	4.49	3.67	2.62	1.48	0.71	0.42	
	RadStd	0.06	0.12	0.21	0.34	0.42	0.49	0.38	0.29	0.14	0.14	0.07	0.04	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	+0.67	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (36 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page